1. System Definition and Gain Selection

Matlab
clear; clc; close all;

K = 200;
s = tf('s');
G = K / (s * (s + 2) * (s + 10));

- **clear; clc; close all;**: Clears the workspace variables, resets the Command Window, and closes any open figure windows.
- **K = 200;**: Sets the open-loop gain to meet the velocity error constant requirement $K_v = 10s^{-1}$:
  $$K_v = \lim_{s \to 0} s \cdot G(s) = \frac{K}{2 \times 10} = \frac{K}{20} = 10 \implies K = 200$$
- **s = tf('s');**: Defines the Laplace variable s for transfer function modeling.
- **G = K / (s * (s + 2) * (s + 10));**: Builds the uncompensated continuous-time plant transfer function.

2. Specification Target Calculations

Matlab
OS = 0.20;
zeta = sqrt((log(OS))^2 / (pi^2 + (log(OS))^2));
PM_req = atan(2 * zeta / sqrt(sqrt(1 + 4*zeta^2) - 2*zeta^2)) * (180/pi);

- **OS = 0.20;**: Defines the maximum allowed percentage overshoot (20%).
  zeta = ...: Computes the required damping ratio $\zeta \approx 0.456$ using the standard relation:

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

- **PM_req = ...**: Computes the exact Phase Margin corresponding to that damping ratio ($\approx 48.1^{\circ}$) using:

$$PM = \tan^{-1}\left(\frac{2\zeta}{\sqrt{\sqrt{1+4\zeta^4}-2\zeta^2}}\right)$$

3. Finding the New Gain Crossover Frequency

Matlab
```
PM_des = PM_req + 6;
target_phase = -180 + PM_des;

w = logspace(-2, 3, 5000);
[mag, phase] = bode(G, w);
mag = squeeze(mag);
phase = squeeze(phase);

[~, idx] = min(abs(phase - target_phase));
w_gc_new = w(idx);
mag_at_w_gc = mag(idx);
```

- **PM_des = PM_req + 6;**: Adds a 6° safety margin to compensate for the small phase lag that the lag compensator inevitably introduces at the crossover frequency.
- **target_phase = -180 + PM_des;**: Sets the target phase on the uncompensated plant's phase curve. Since $PM = 180^{\circ} + \angle G(j\omega)$, the required phase is $-180^{\circ} + PM_{des}$.
- **w = logspace(-2, 3, 5000);**: Generates 5,000 logarithmic frequency points from $0.01 rad/s$ to $1000 rad/s$.
- **[mag, phase] = bode(G, w); ... squeeze(...);**: Calculates the magnitude and phase response of the plant across frequency vector w, flattening the resulting 3D arrays into standard 1D vectors.
- **[~, idx] = min(abs(phase - target_phase));**: Finds the index where the plant's phase is closest to target_phase.
- **w_gc_new & mag_at_w_gc**: Extracts the new gain crossover frequency ($\omega_{gc,new}$) and the plant's magnitude at that exact frequency.

4. Lag Compensator Design

Matlab
```
beta = mag_at_w_gc;
```

w_z = 0.1 * w_gc_new;
w_p = w_z / beta;

D_lag = (1 + s / w_z) / (1 + s / w_p);
G_comp = D_lag * G;

- **beta = mag_at_w_gc;**: The lag compensator provides a high-frequency attenuation of $1/\beta$. To make $\omega_{gc,new}$ the new 0 dB crossover frequency, β must equal the plant's magnitude at that point.
- **w_z = 0.1 * w_gc_new;**: Places the zero corner frequency (ω_z) one decade below $\omega_{gc,new}$ so that the compensator's phase lag is mostly finished by the time the system reaches the crossover frequency.
- **w_p = w_z / beta;**: Computes the pole corner frequency (ω_p), where $\omega_p = \frac{\omega_z}{\beta}$.
  D_lag = ...: Forms the lag compensator transfer function:

$$D(s) = \frac{1+\frac{s}{\omega_z}}{1+\frac{s}{\omega_p}} = \frac{s+\omega_z}{s+\omega_p} \cdot \frac{1}{\beta}$$

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- **G_comp = D_lag * G;**: Cascades the compensator with the plant to form the open-loop compensated system.

5. Margin and Closed-Loop Analysis

Matlab
[Gm, Pm, Wcg, Wcp] = margin(G_comp);
Gm_dB = 20 * log10(Gm);

T_comp = feedback(G_comp, 1);
[r, p, ~] = residue(T_comp.Numerator{1}, T_comp.Denominator{1});

- **margin(G_comp);**: Computes the stability margins (Gain Margin Gm in absolute ratio, Phase Margin Pm in degrees, phase crossover frequency Wcg, and gain crossover frequency Wcp).
- **Gm_dB = 20 * log10(Gm);**: Converts the Gain Margin to decibels (verifying the $\geq 20dB$ requirement).
  T_comp = feedback(G_comp, 1);: Computes the closed-loop transfer function with unity negative feedback:

$$T(s) = \frac{G_{comp}(s)}{1+G_{comp}(s)}$$

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- **residue(...)**: Calculates the poles p and residues r of the closed-loop system via partial fraction expansion to examine pole dominance.

6. Value Display and Plotting

Matlab
beta
w_z
w_p
Pm
Gm_dB
p
r

```
figure;
subplot(2, 1, 1);
bode(G, D_lag, G_comp);
legend('G', 'D', 'G_{comp}');
grid on;

subplot(2, 1, 2);
step(T_comp);
grid on;
```

- **Variables printed without semicolons**: Outputs the key design parameters (β, upper/lower corner frequencies ω_z and ω_p, Phase Margin, Gain Margin in dB, closed-loop poles, and residues) directly to the Command Window.
- **subplot(2, 1, 1)**: Plots the open-loop Bode responses of the uncompensated plant G, lag compensator D_lag, and combined compensated system G_comp together.
- **subplot(2, 1, 2)**: Plots the closed-loop unit step response of T_comp to visually confirm transient response characteristics (rise time, overshoot, and settling behavior).